ATTACK DEMONSTRATION
SECURE SYSTEM

Defense-in-Depth: Same Attack, Zero Compromise

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SECURE SYSTEM: Pi #6 — 10.30.0.25 (VLAN30)

⚡ ATTACKER — Suricata Laptop (192.168.50.191 VLAN2)

```
$ python3 attack_spoof.py --dual --pressure 0.0 --rate 0.1
```

Sends fake pressure readings: 0.0 PSI at 10 Hz (one request every 0.1 seconds)

```
HTTP POST /sensor → {"pressure": 0.0, "source": "attacker"}
```

⛔ BLOCKED

Pi #6 — SECURE BALLOON CONTROL SYSTEM | 10.30.0.25 | Raspberry Pi 5 | Flask API : 5000

Flask API /sensor

- ✓ RATE LIMITING (2 Hz max)
- ✓ INPUT VALIDATION (0–100 PSI)
- ✓ SENSOR CROSS-CHECK
- ✓ ANOMALY DETECTION
- ✓ WAZUH SIEM LOGGING

Validates ALL data

Validation Logic

1. Network claim: 0.0 PSI
2. I2C sensor: 25.0 PSI
3. Mismatch: |0.0–25.0| = 25.0 PSI
4. Threshold: 25.0 > 1.0 PSI FAIL!
- ✓ Attack DETECTED
- ✓ Logged to Wazuh SIEM
- ✓ Network data REJECTED

Control Loop

1. Uses I2C: 25.0 PSI (TRUSTED)
2. Target: 25 PSI
3. Deficit: 25 – 25 = 0 PSI
4. Decision: MAINTAIN ✓
- ✓ Fallback to I2C active
- ✓ Attack has ZERO effect

Relay Control (Sequent HATs)

8-Relay HAT (I2C 0x27):

```
Relay 2 — Inflate: CLOSED
Relay 3 — Vent: CLOSED
Relay 4 — Relief: CLOSED
```

40A 3-Relay HAT (I2C 0x60):

```
Relay 3 — Pump: OFF ✓
```

PHYSICAL PRESSURE VESSEL (1L Bottle)

Real Pressure: 25.0 PSI (SAFE — maintained at target)

System correctly reads I2C sensor: 25.0 PSI (TRUSTED)

Network spoofing attack: BLOCKED — no effect on control loop

Pressure remains stable at 25 PSI

Safety Margin: 14+ PSI below burst threshold (39–71 PSI)

✓ PRESSURE REMAINS SAFE ✓

✓ SECURITY CONTROLS

- ✓ Rate limiting (2 Hz max)
- ✓ Input validation
- ✓ Sensor cross-check
- ✓ Anomaly detection
- ✓ SIEM logging (Wazuh)

Result: 0% attack success

DEFENSE PROGRESSION: ATTACK BLOCKED

T = 0 sec	T = 5 sec	T = 10 sec	T = 15 sec
NORMAL OPERATION	**ATTACK DETECTED**	**ATTACK BLOCKED**	**SYSTEM SAFE**
Real Pressure: 25.0 PSI	Real Pressure: 25.0 PSI	Real Pressure: 25.0 PSI	Real Pressure: 25.0 PSI
Network Reading: 25.0 PSI	Network Reading: 0.0 PSI (FAKE!)	Network Reading: 0.0 PSI (FAKE!)	Network Reading: 0.0 PSI (FAKE!)
I2C Sensor: 25.0 PSI (TRUSTED)	I2C Sensor: 25.0 PSI (TRUSTED)	I2C Sensor: 25.0 PSI (TRUSTED)	I2C Sensor: 25.0 PSI (TRUSTED)
Pump Status: OFF (at target)	Pump Status: OFF (maintained)	Pump Status: OFF (maintained)	Pump Status: OFF (maintained)
Detection: ✓ No anomaly	Detection: ✓ DETECTED!	Detection: ✓ BLOCKED!	Detection: ✓ Still safe!
System safe	Mismatch: 25 PSI	Mismatch: 25 PSI	Mismatch: 25 PSI
Attack Traffic: None	Attack Traffic: 10 req/sec All rejected	Attack Traffic: 10 req/sec All rejected	Attack Traffic: 10 req/sec All rejected

DEFENSE SUCCESS METRICS

0%	10/sec	150+	< 1s
attack success	attack rate	rejected	detection
every request blocked	all rejected	all logged to Wazuh	real-time response

DEFENSE OUTCOME: ATTACK FAILED

PHYSICAL SAFETY

BOTTLE STATUS (T = 15 sec)

Real Pressure: 25.0 PSI

System Belief: 25.0 PSI (I2C trusted)

Safe Limit: 25 PSI | Status: AT TARGET ✓

Safety Margin: 14+ PSI below burst

SYSTEM INTEGRITY

- ✓ Bottle safe | ✓ All valves nominal
- ✓ Sensors accurate
- ✓ Emergency shutoff ready

REAL-WORLD IMPACT

- ✓ Medical device protected
- ✓ Patient safety maintained | Attack fully mitigated

DETECTION & RESPONSE

SECURITY MONITORING

- ✓ Sensor mismatch detected (25 PSI delta)
- ✓ Anomaly threshold triggered
- ✓ All attacks logged to Wazuh SIEM
- ✓ Real-time alerts generated

DEFENSE MECHANISMS

1. I2C cross-check: PASSED
2. Rate-of-change: NORMAL
3. Range validation: IN BOUNDS

WAZUH SIEM LOGGED

- ✓ 150+ rejections | Attacker: 192.168.50.191
- ✓ Forensic data preserved | Alert: HIGH

150+	150+	< 1s	25.0
attacks blocked	detections	detection	PSI maintained
0% success rate	100% logged to Wazuh	real-time response	at target (SAFE)